

Syllabus on Reinforcement Learning for Neuroscience

This document presents a comprehensive 10-session syllabus for the reinforcement learning study group. The curriculum is structured to build from foundational principles to the cutting edge of research, with a persistent focus on the links between reinforcement learning (RL) algorithms and neural substrates.

Each session is designed with two parallel tracks to serve the entire group:

1. **Conceptual & Neuroscience Track (Accessible):** Provides an intuitive, accessible explanation of the core RL concept and its direct application to a key question in neuroscience.
2. **Mathematical Deep Dive (Advanced):** Provides the formal mathematical definitions, derivations, and proofs for participants comfortable with maths

Some key references:

A classic textbook:

<https://web.stanford.edu/class/psych209/Readings/SuttonBartoIPRLBook2ndEd.pdf>

A complete overview of modern RL: <https://arxiv.org/abs/2412.05265>

An online course by University of Alberta:

<https://www.coursera.org/specializations/reinforcement-learning>

Table 1: Syllabus Overview and Roadmap

Session	Core RL Topic (Accessible)	Key Neuroscience Connection (Systems/Comp)	Core Reading(s)	Mathematical Deep Dive (Advanced)
1	Agent, State, Action, Reward, Exploration. Policy (π), Value Function (V , Q), and the Bellman Equation.	Marr's Levels: RL as a "normative" theory of reward-guided behavior. ¹ VMPFC and OFC encoding subjective value. ⁶	Sutton & Barto, Ch. 1. ² Niv (2009) Review. ¹	Markov Decision Processes (MDPs): Formal tuple (S, A, T, R, γ) and the Markov Property. ³ Bellman Optimality: Derivation of the

				optimality equation ⁸ and solution via Value/Policy Iteration. [4, 9] Contraction mapping proof.[10]
2	Model-Free Learning: Monte Carlo (MC) vs. Temporal Difference (TD) "Bootstrapping". [11, 12]	The Seminal Link: Phasic dopamine as a TD Reward Prediction Error (RPE). [13, 14]	Schultz, Dayan, & Montague (1997). [15, 16, 17]	TD(0) Algorithm: Bias-variance trade-off. The "dopamine asymmetry" problem (low baseline firing).[18, 19, 20]
3	TD Control: On-Policy (SARSA) vs. Off-Policy (Q-Learning). [21, 22, 23]	Behavioral Implications: "Cliff-world" example. ²¹ Risk-aversion (SARSA) vs. Optimality (Q-Learning). ²³	Sutton & Barto, Ch. 6.	Convergence Proof: Stochastic approximation conditions for Q-learning convergence. [25, 26]
4	Scaling Up: The problem of large state spaces and the need for Function Approximation. ²⁷	DRL as a Model: Using Deep Q-Networks (DQN) as "virtual subjects" to model high-dimensional tasks. [28, 29, 30]	Mnih et al. (2015) Nature paper.	The "Deadly Triad": Why off-policy + bootstrapping + function approximation diverges. ³¹ DQN solutions: Experience Replay & Target Networks. ²⁷
5	Policy Gradients: Learning a policy $\pi(a s; \theta)$ directly. Advantages for continuous action	Motor Control: Modeling continuous motor commands (e.g., reaching) as a direct policy	Sutton & Barto, Ch. 13.	The Policy Gradient Theorem: Derivation of the "log-derivative" trick. [36, 37] The

	spaces. [32, 33]	optimization. ³⁴		REINFORCE algorithm [38] and variance reduction with baselines. ³⁹
6	Actor-Critic Methods: Combining value functions (Critic) and policy functions (Actor). [40, 41]	The Basal Ganglia: Canonical model. Dorsal Striatum (Actor) and VTA/Dopamine (Critic). ⁴²	Joel et al. (2002) Review. [42, 45, 46]	Advantage Actor-Critic (A2C): Derivation. This as a TD-based, low-variance policy gradient algorithm. ³⁹
7	Model-Based RL: Model-Free (Habit) vs. Model-Based (Goal-Directed) dichotomy.[47, 48, 49]	Dual Brain Systems: DLS (Habit) vs. PFC/DMS (Goal). ⁵⁰ Hippocampus as a "Cognitive Map". ⁵³	Daw & Doya (2012) Review.[52]	The Successor Representation (SR): A bridge between MF and MB. ⁵⁵ The Dyna-Q Algorithm. [57, 58]
8	RL in humans:	Predictive representations: a general principle for memory organisation?	Momennejad, ..., Gershman 2017 [72], Russek, Momennejad et al 2017 [73],	The Dyna-SR algorithm. Python code: https://github.com/idamomen/predictive_representations
9	Uncertainty: Partially Observable MDPs (POMDPs) and "Belief States". ⁵⁹	Evidence Accumulation: POMDPs as a normative model for decision-making. ⁶¹ LIP cortex encoding the "belief state". ⁶²	Rao (2010) Review.[63]	The POMDP Value Function: Bayesian belief updates. ⁶² The Bellman equation over the (continuous) belief simplex.
10	The Virtuous Circle: The modern frontier of AI and neuroscience. ⁶⁴	1. Advanced Dopamine: Distributional RL [65, 66] & Sensory PEs.[67, 68] 2.	Hassabis et al. (2017) Neuron. ⁶⁴ Dabney et al. (2020) Nature. ²⁹	Analysis of Distributional Bellman operator. Math of episodic control (non-

		Neuro-inspired AI: Episodic Memory ⁶⁹ & Attention. ⁷¹		parametric) and attention (soft vs. hard).
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